

Wave Chuang

Capital project leader who bridges chemical engineering know-how with strong execution—delivering high-purity gas and chemical supply systems for semiconductor fabs on time, on budget, and in compliance.

Capital Project Leader | Semiconductor & Chemical Industries | \$1.2B+ Delivered | High-Purity Production Systems

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PMP-certified project manager and **certified industrial engineer** with **12+ years** delivering capital projects in chemicals, specialty materials, and the semiconductor supply chain. Hands-on across the **full project lifecycle**—from **FEED** and feasibility through **EPC** build and **on-stream commissioning**. Designed and delivered **skid-mounted high-purity specialty chemical production systems** that supply **TSMC's advanced-node fabs**. Led **cross-national engineering teams** across US, Korea, Germany, and Taiwan. Track record: **on time, on budget**, with **zero change orders** and **zero safety incidents**. Strong in **regulatory and environmental compliance**—EIA, air/water permits, occupancy—and in **HAZOP**, risk assessment, and **change management**. **Fluent in English and Mandarin**. Known for meeting the high bar on purity and quality that semiconductor processes demand.

Key Skills

Chemical Eng. Expertise

Certified Industrial Engineer with end-to-end project delivery in chemical and specialty materials territory.

CAPEX & investment

Feasibility, FEED, cost estimates for FID; data for global investment committees.

Risk Management(HAZop)

Proactively assess risks, implement mitigation strategies, and ensure project resilience under uncertainty.

AI-enabled delivery

AI to turn requirements/data into visuals and workflows; alignment and transparency.

Regulatory & environmental

EIA, air/water permits, occupancy permitting.

Cross-national leadership

Led cross-functional teams (US, Korea, Germany, Taiwan) in matrix orgs; milestones and quality on track.

Work Experience

Senior Software Project Manager - Foxxcon Dec 2023 - Present

- Led development and integration of an EV TSP platform and HPC firmware (SOVD diagnostics, OTA) for CEER, Saudi Arabia's first EV brand.
- Managed scope, schedule, and risk across a multi-supplier program with clear milestones and change control.
- Used AI-enabled workflows to turn requirement docs into traceable task lists and improve alignment and transparency.

Capital Project Manager - Merck Group Sep 2022- Dec 2023

- Ran a **€12M** capital project (state-of-the-art fine chemical line supplying TSMC advanced-node processes) under the Level-UP program.
- Led cross-national, cross-functional teams in a global matrix; hit schedule and quality targets.
- Ran FEED studies for scope, cost, and feasibility; built roadmaps, resource plans, and risk registers with mitigations.
- Delivered cost estimates and financial analysis so the Global Capital Investment Committee could make a solid Final Investment Decision (FID).
- Oversaw construction and occupancy permitting and environmental assessments for local permits.

 **Nominated for Merck Taiwan Q-Star Award (2023)** for high-impact innovation on the process line.

Project Development Manager - Taiwan Cube Energy Jan 2022 - May 2022

- Led prep and submission of government tender proposals; kept technical scope aligned with regulations.
- Did market and competitor analysis to find differentiators and strengthen bids.

Advanced Project Manager - Formosa Chemicals & Fibre Dec 2011 – Jan 2022

- Managed the **\$1.2B Sunshine Project** greenfield petrochemical complex in Louisiana—from process R&D through FEED and permitting.
- Delivered EPC with zero change orders and on-time completion; zero casualties and zero near-misses on site.
- Ran feasibility studies that secured Final Investment Decisions from senior leadership.
- Managed EIA and emission control; coordinated land-use permitting with the U.S. Army Corps of Engineers.
- Led a cross-national team on a **\$30M** power plant FGD project in Vietnam—from process development through remote-site commissioning.

Education and Certification



National Taiwan University
Chemical engineering
2005 - 2010

- Project Management Professional (PMP), PMI
- Certified Industrial Engineer, CIIE
- Dedicated Wastewater Treatment Specialist (Class A), EPA
- Dedicated Air Pollution Control Specialist (Class A), EPA
- Waste Disposal Technician (Class A), EPA
- Microsoft Certified: Azure Administrator Associate

Major Project List

Specialty Production Line Project, TW (€3.5M)

Designed and delivered a fully self-developed, state-of-the-art skid-mounted production line supplying TSMC advanced-node materials, including high-purity electronic specialty chemicals and the associated supply system design and construction, from R&D concept to on-stream.

SunShine Project, Louisiana USA (\$1.2B):

Managed FEL-1/2/3 stages for a greenfield petrochemical complex; secured air and land-use permits.

Power Plant SCR Project, TW (\$0.7M)

Implemented NOx control technology using aqueous ammonia.

Fine Chemical Facility Project, TW (€12M)

Conducted FEED, cost analysis, and execution roadmap for a new fine-chemical production line supplying TSMC advanced-node technologies

Power Plant FGD Project, VN (\$30M)

Developed advanced flue-gas desulfurization processes and led a cross-national engineering team.

Turbine Installation, Chiayi Taiwan

Served as Construction Manager ensuring seamless multi-disciplinary coordination and on-schedule delivery.